

# Khalil Lagha

**M2 student in Intelligent Systems for Automotive & Aerospace (SIAA) — embedded systems, control & power electronics · Université Paris-Saclay · Work-study via CFA EVE**

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## SUMMARY

M2 student in Intelligent Systems for Automotive & Aerospace (SIAA, Université Paris-Saclay, **work-study via CFA EVE**), **ranked 2nd of the M1 EEA cohort (15.43/20)**. Specialized in **control and embedded systems**: field-oriented control (FOC), PID controllers, observers, **Kalman filter**, **Matlab/Simulink** validation (specification → implementation → test) then real-hardware testing — across **five internships** from the lab (GIPSA-lab) to industrial sites. Target fields: **EV / powertrain, embedded systems, applied AI**. **Seeking: work-study contract (alternance) from September 2026 — 2 days company / 3 days university.**

## EXPERIENCE

**Research Intern — GIPSA-lab, Grenoble** | 04–07/2026 · 3.5 months

- Modeled a **PEMFC fuel cell** in Matlab/Simulink — **specification → implementation → test** approach; achieved **6 %** dynamic error and **1 %** static error against real measurements.
- Tuned a **DC-DC boost converter PI regulation loop** to optimize efficiency; validated first in simulation, then on a real test bench (tests, measurements, documentation).

**Intern — SLB / Schlumberger** | 07/2025 · 1 month

- Authored technical summaries in **English** (directional drilling) in an international industrial environment.

**Power Electronics & Automation Intern — EURL Lagha** | 01/2024 · 15 days + 09/2024 · 1 month

- Contributed to the electronics of real industrial equipment: a **DC-DC buck converter** (50 V → 0–48 V adjustable, **25 A**) and **4 transformer-rectifiers** (100 V DC / 80 A ≈ 8 kW, ripple < 5 %).
- Co-designed the supervision HMI (**Siemens TIA Portal**, Ladder PLC); configured industrial control applications.
- Analyzed a customer specification and sized the corresponding transformer; produced the electrical schematics in **EPLAN** — direct interface with manufacturing.

**Power Grid Intern — Sonelgaz** | 03/2024 · 15 days · observation

- Analyzed the full generation → distribution chain (**400/220 kV → 60 kV → 30/10 kV → 400/230 V**) and substation architecture; surveyed an MV/LV substation during a technical site visit.

## EDUCATION

**M2 SIAA — Intelligent Systems for Automotive & Aerospace · Université Paris-Saclay (Évry) · CFA EVE** | 09/2026 · admitted · work-study

**M1 EEA (EECS) · Université Grenoble Alpes** | 2025 – 2026

**Ranked 2nd of the cohort** · annual average **15.43/20**.

**Engineering programme in Electrical Engineering · École Nationale Polytechnique, Algiers** | 2021 – 2025

4 of 5 years completed · moved to France for the master's · prep classes: among the top-ranked.

**Baccalauréat in Mathematics** | 2021

Highest honours (Très Bien) — **16.93/20**.

## PROJECTS

**Field-oriented control (FOC) of an induction motor — Matlab/Simulink** **FLAGSHIP**

- Implemented Park transforms, PID controllers, observers and a **Kalman filter**; tuned the **cascaded current- and speed-regulation loops** — the core of electric-vehicle traction.

**PWM generator with adjustable frequency and duty cycle — 555 timer + D flip-flops**

- Designed a **PWM** generator with **adjustable frequency and duty cycle**; simulated in **Proteus** then **built and fine-tuned on real hardware** — a complete electronics-diagnostics workflow.

**Neural network from scratch (MLP) — Matlab · open source**

- Hand-coded a multilayer perceptron and its **backpropagation** without any toolbox; debugged, tested and **published under the MIT license** — [github.com/KhalilEllahLAGHA/mlp-backpropagation-matlab](https://github.com/KhalilEllahLAGHA/mlp-backpropagation-matlab).

**Electrical machine sizing — Python · team**

- Built a **Python (OOP, PyQt)** application — from specification to automatic parameter generation; finite-element validation (**FEMM**).

Also: cascade control of a DC motor — three-phase thyristor bridge (Simulink); Siemens Step7 automation (HMI + Ladder); Arduino line-follower robot (Poly Maze competition); power grid analysis (load flow, 3–118 buses, Matlab).

## SKILLS

**Control engineering:** PID, LQR, observers, Kalman filter · field-oriented control (FOC) · cascade control · regulation-loop tuning · system identification

**Embedded systems & programming:** C/C++, Python (OOP, PyQt), Arduino (IDE + firmware) · hands-on electronics, from schematic to real circuit (Proteus, LTspice)

**Industrial automation:** Siemens PLC (TIA Portal, Step7 — Ladder, SCL, FBD, SFC) · HMI/SCADA · Grafcet

**AI:** neural networks (backpropagation), genetic algorithms, data fusion, fuzzy logic

**Power electronics:** DC-DC conversion (buck, boost), AC-DC, DC-AC, AC-AC · PWM · power electronics for machine control

## SOFTWARE

Matlab/Simulink · Python (OOP, PyQt) · C/C++ · Arduino · Proteus · LTspice · Siemens TIA Portal · Step7 (Ladder/SCL) · EPLAN · AutoCAD Electrical · FEMM · SolidWorks · Git/GitHub

## SOFT SKILLS & ACTIVITIES

Fast learner · comfortable with new tools & software · problem solving · rigorous · organized — IEEE Student Branch ENP · VIC (ENP)

## CERTIFICATIONS

SLB — NEST & SIPP Level 2 (07/2025) — HSE, industrial operations safety

## LANGUAGES

French — C1 · English — C2